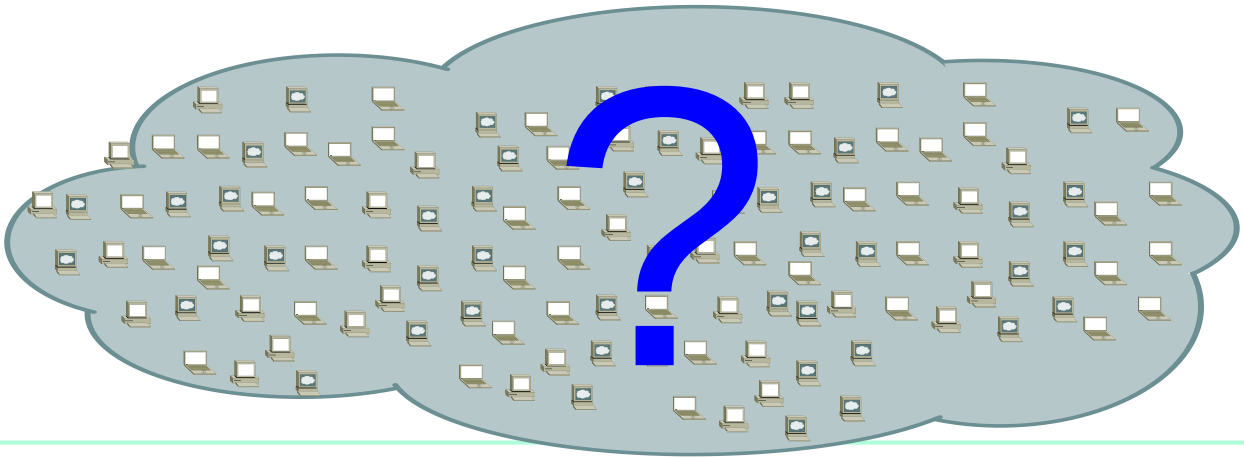
# Router Introduction

# Router Introduction

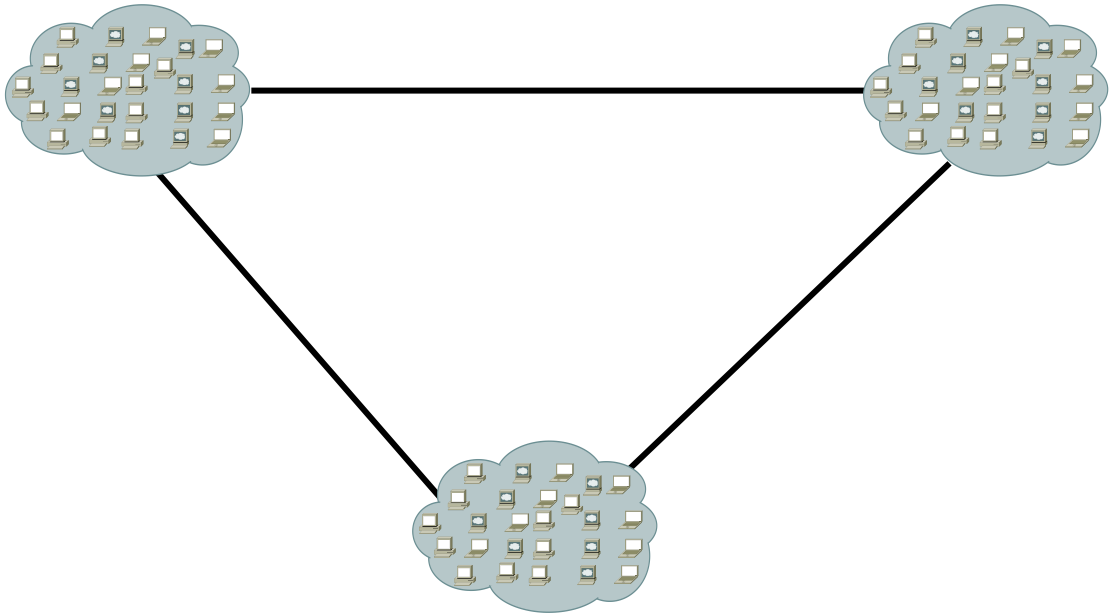
# Router

- With a network that has hundreds or even thousands of computers linked together, there has to be some agreed-upon way for those devices to address one another and communicate.
- As a network grows larger, it is not feasible for each computer to keep track of the individual address of every other computer on the network.
- There must be some scheme for reducing the amount of information, each computer has to hold locally in order to communicate with every other computer.



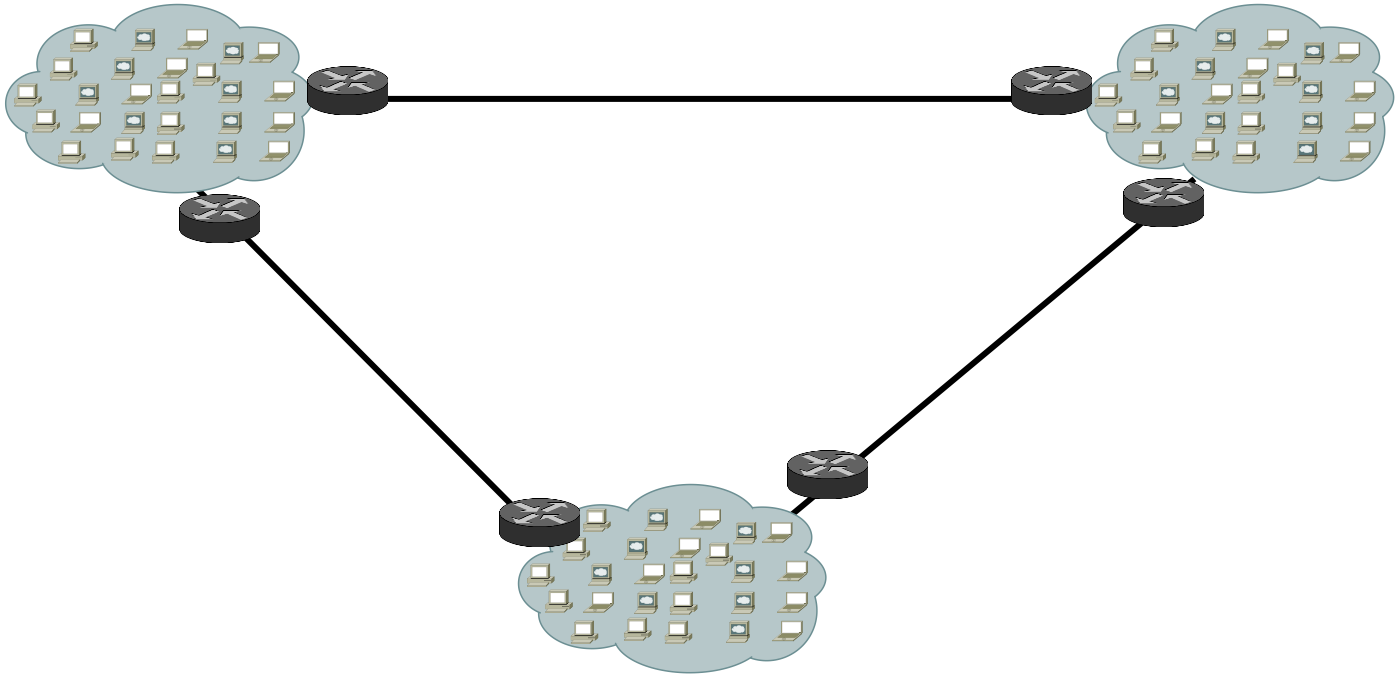
# Router (Cont.)

- The scheme used to involve splitting an large network into many separate but connected networks.



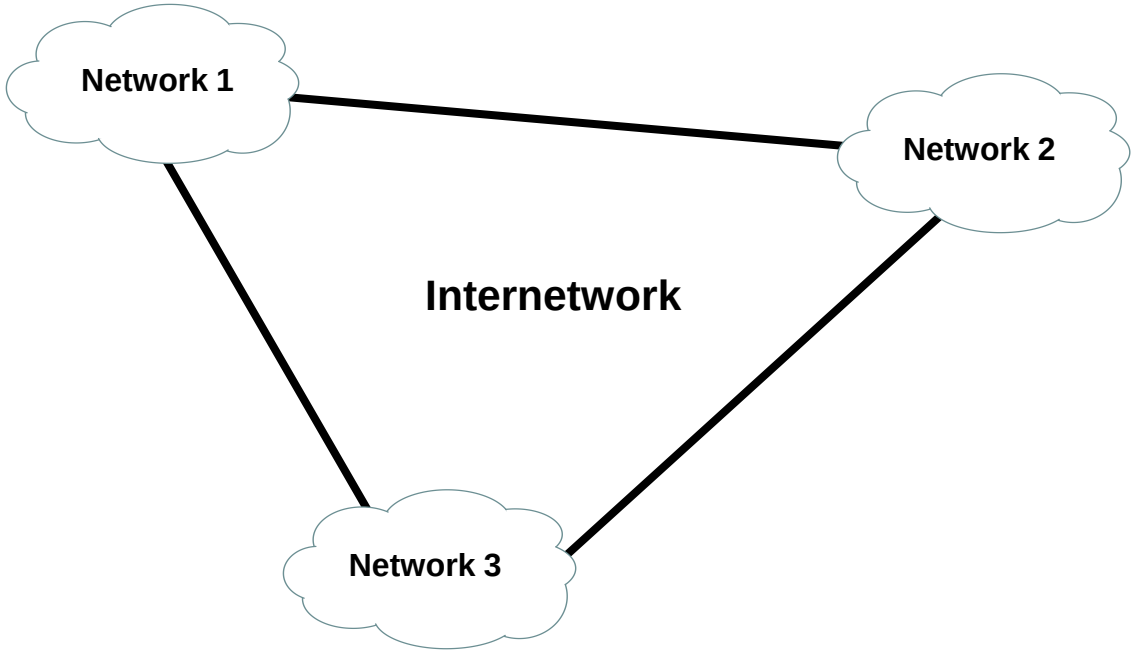
# Router (Cont.)

- The job of keeping track of these discrete networks is then given to specialized computers called routers.



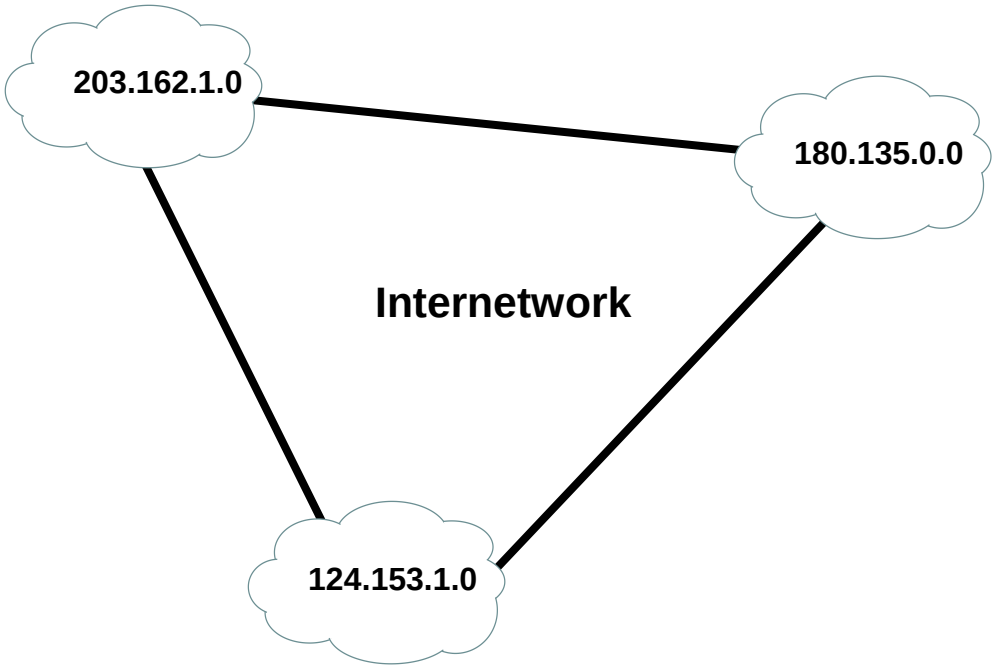
# Internetwork

- An *Internetwork* is an accepted networking industry term for a set of interconnected networks.



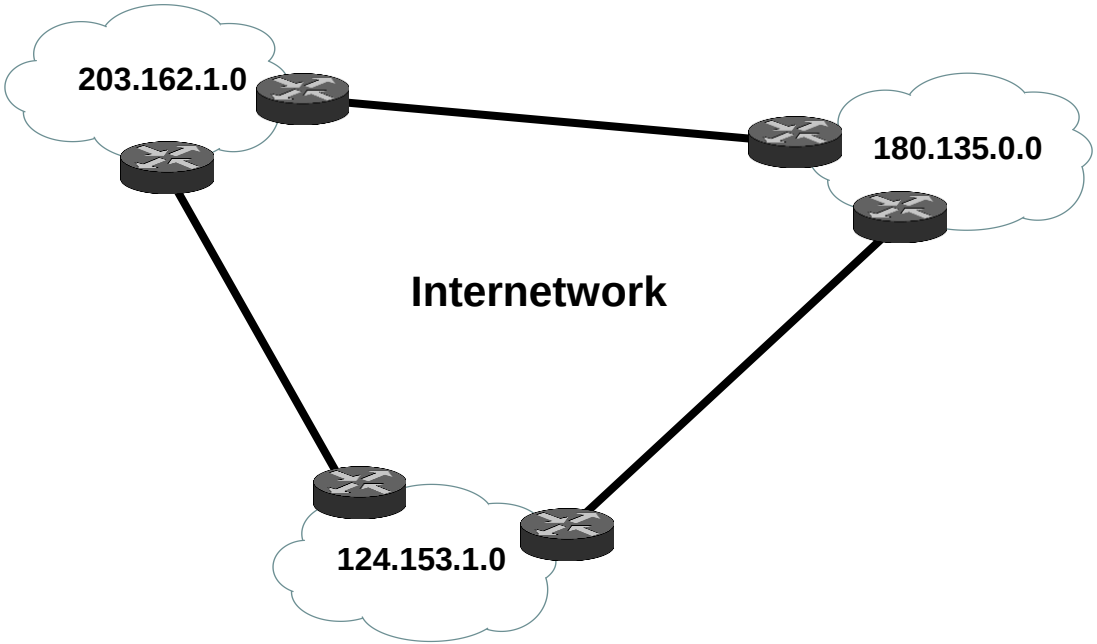
# Internetwork (Cont.)

- Each individual network will have its own network number that must be unique for that internetwork.



# Internetwork (Cont.)

- *Routers* direct traffic through an internetwork, based on information learned from network protocols.



# Router's Function

- Routers are used to route packets from one network number to another.
- There are two things to consider with router:
  - You cannot configure the same network number on more than one interface on a router.
  - A router does not forward broadcasts by default.



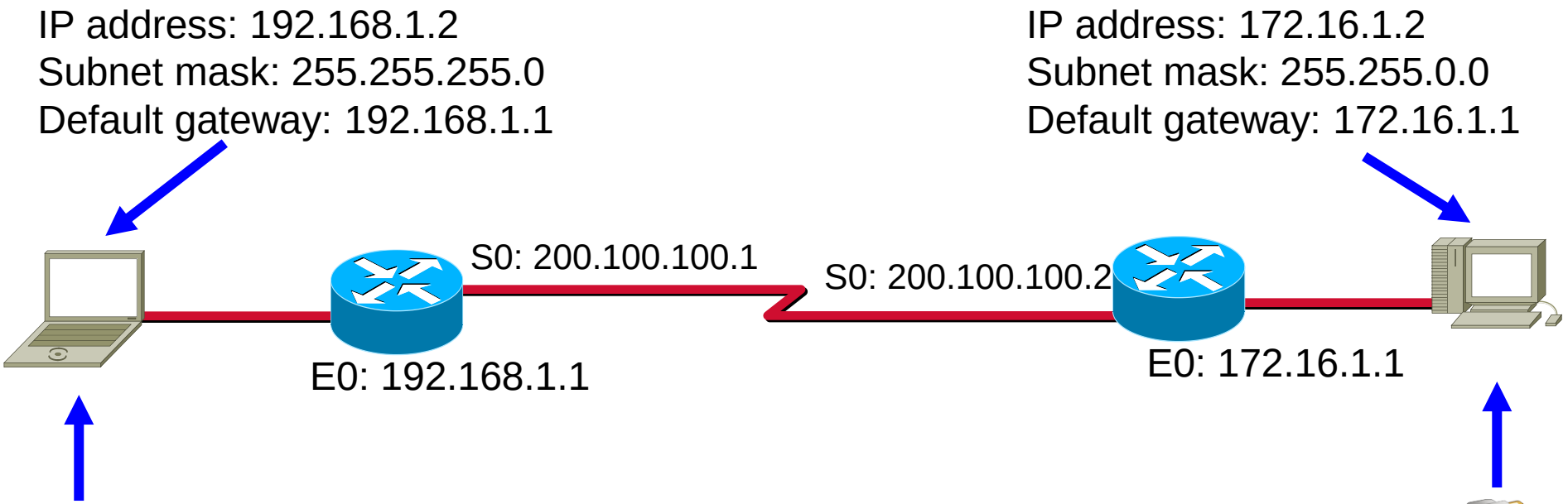
# Network Communication

- Let's take a look at the way Layer 3 network devices communication with each other.
- A typical workstation (a PC running a popular TCP/IP stack, for example) will require some manual configuration before it can operate on a TCP/IP network.



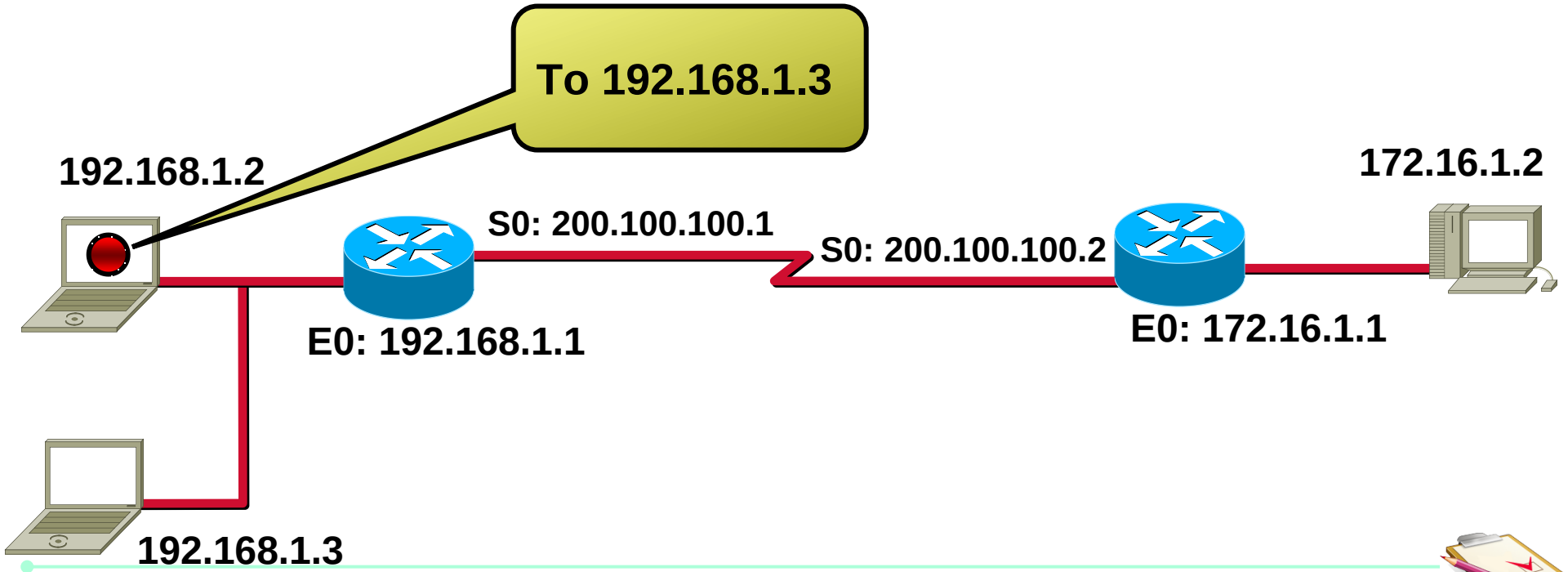
# Network Communication (Cont.)

- At a minimum, you will have to configure an IP address, a subnet mask, and a default gateway.



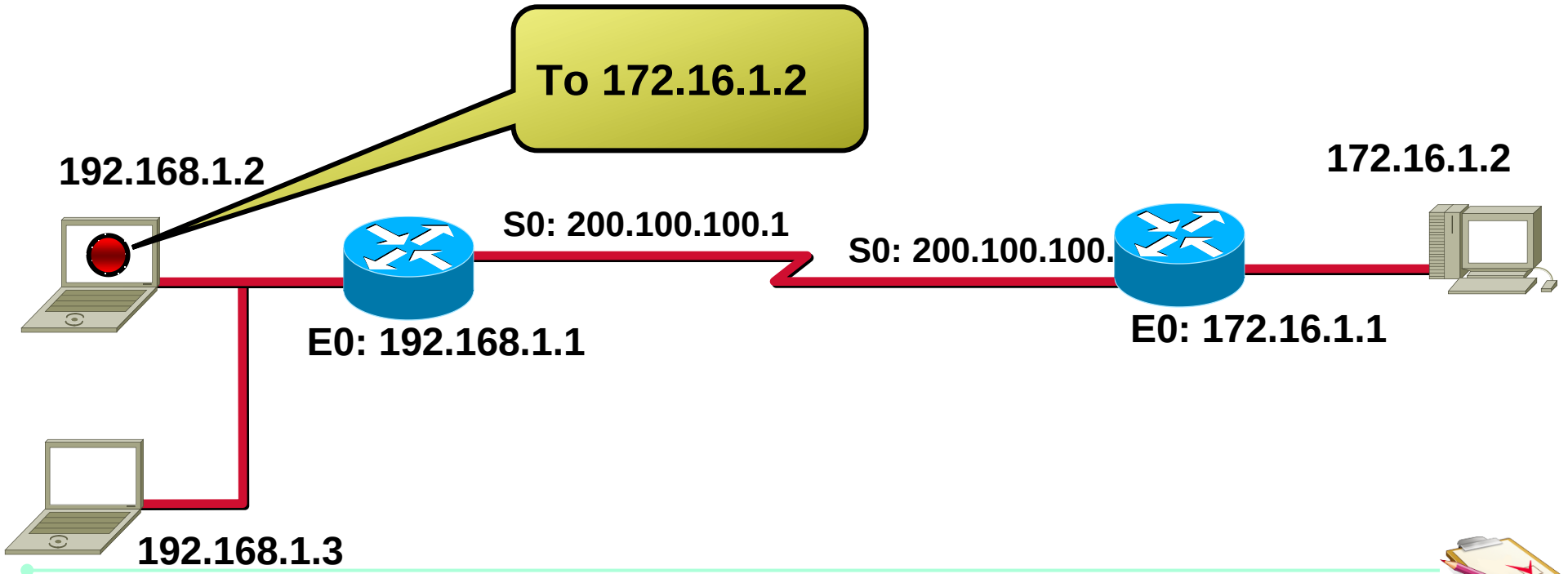
# Network Communication (Cont.)

- The routing decisions of a workstation configured in this manner are simple.
- If the workstation has to send a packet to another machine that is on the same network number, the packet is sent directly to the destination machine.



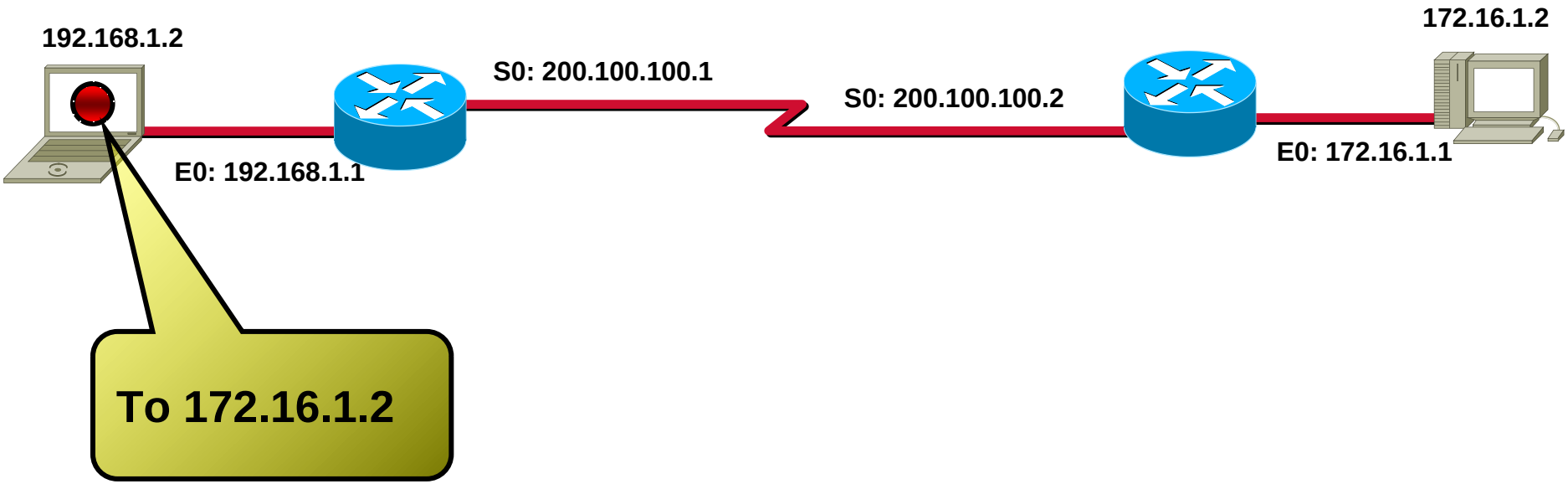
# Network Communication (Cont.)

- If the destination is on a different network number, the packet is forwarded to the default gateway for routing through the internetwork and on to the final destination.



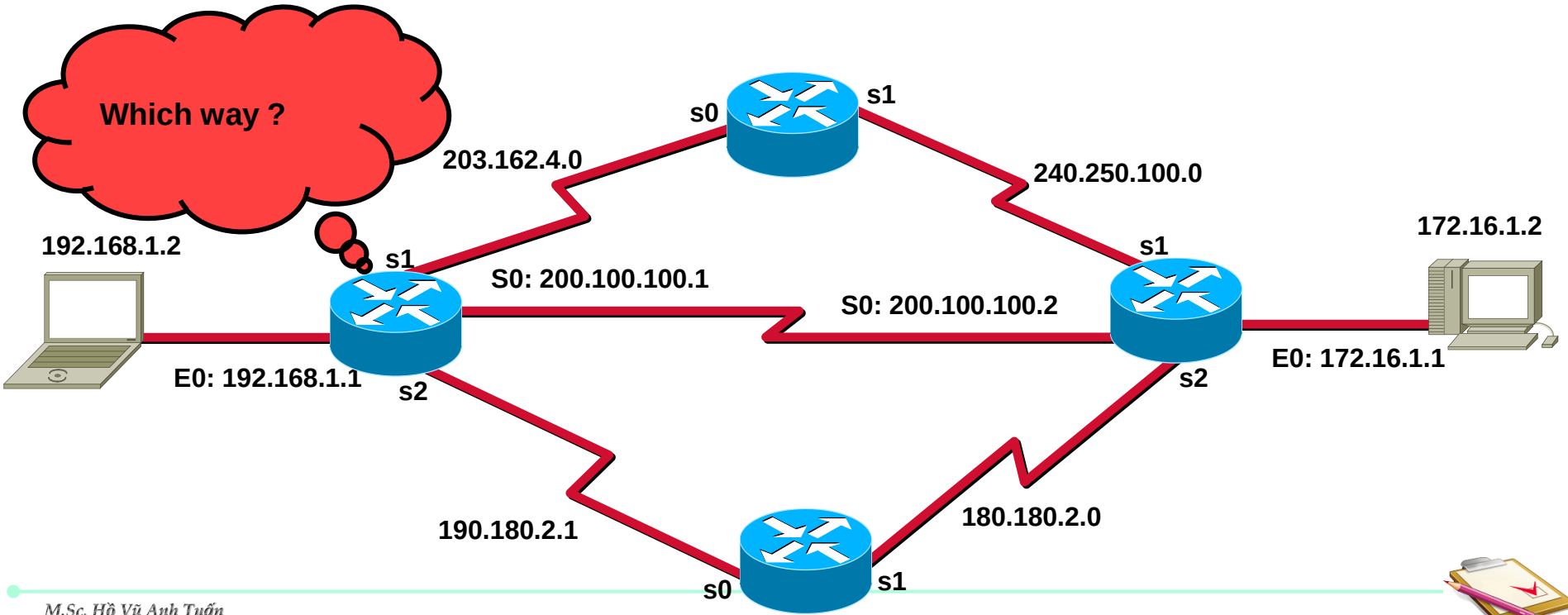
# Network Communication (Cont.)

- Routers make more complex decisions.



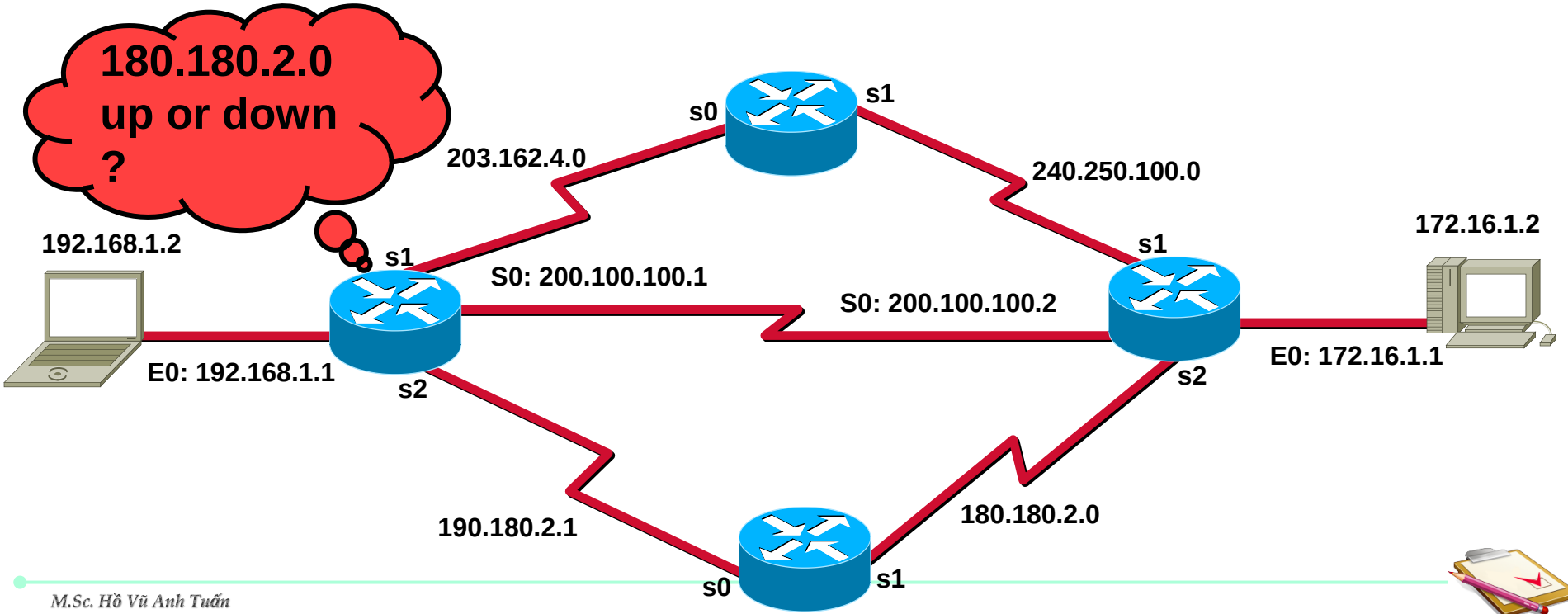
# Network Communication (Cont.)

- They must know how to get to all other network numbers on the internetwork and the best way to route the packets.



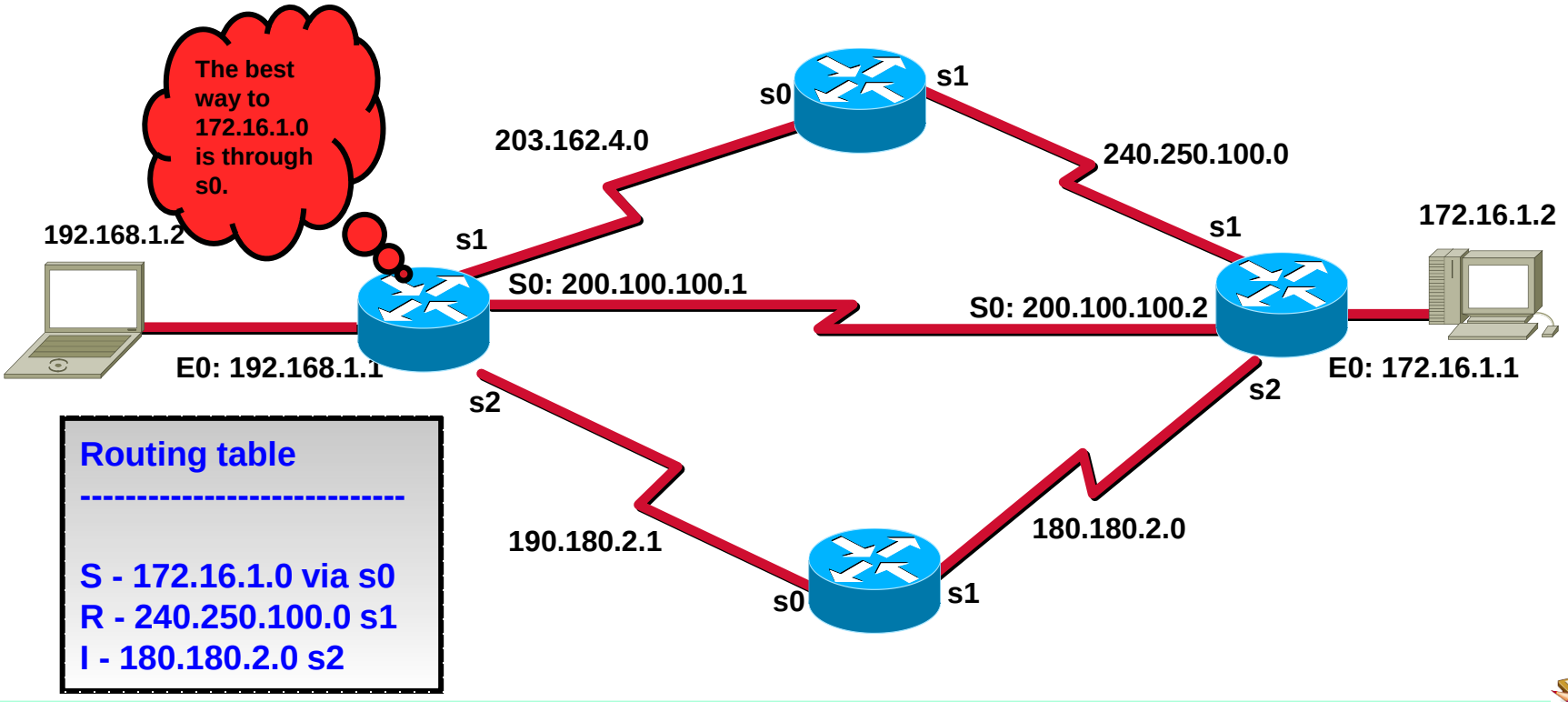
# Network Communication (Cont.)

- And they need to keep track of an internetwork topology that is constantly changing due to equipment or other failures.



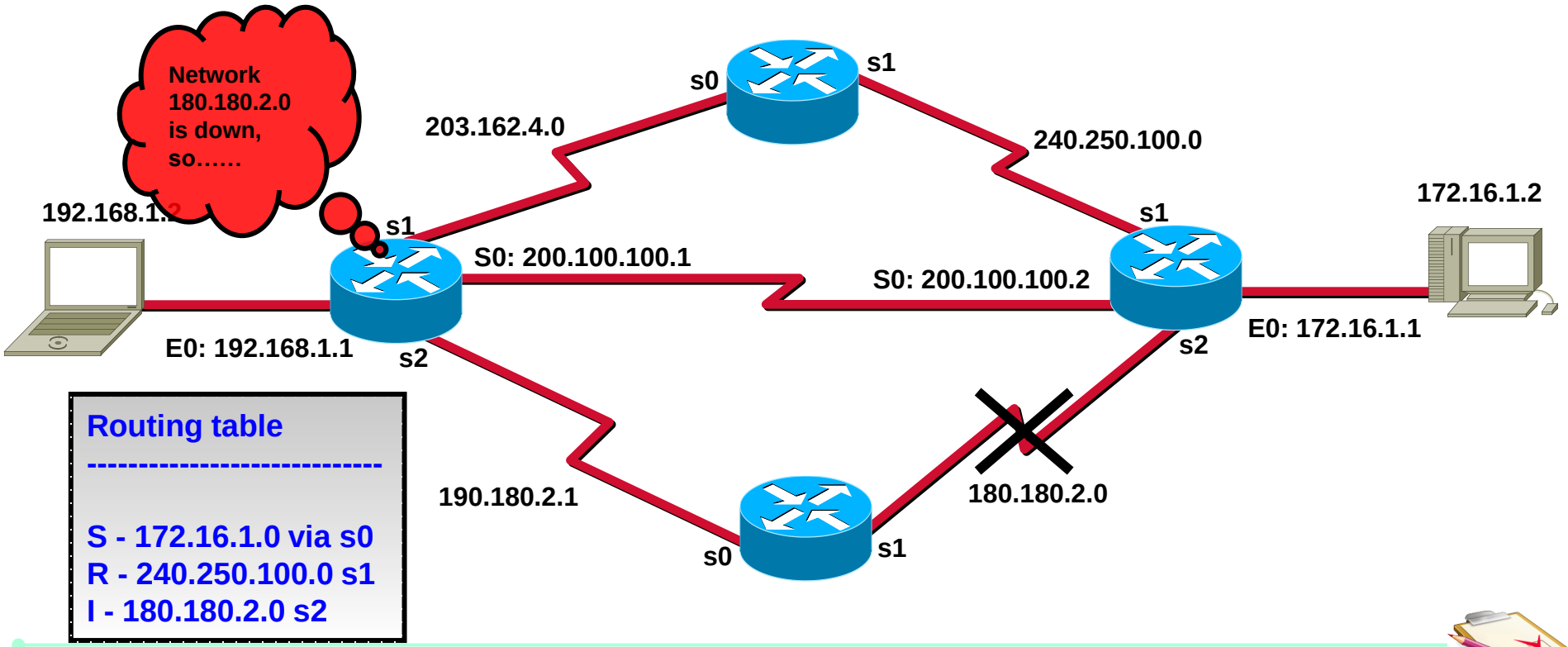
# Routing Table

- To accomplish these responsibilities, a router maintains a *routing table*, which lists all the known network numbers and how to get to them.



# Routing Table (Cont.)

- Routers also use routing protocols that keep the routing table accurate for a changing internetwork.



# Router's Component

A Cisco router is similar to and different from other computers on an internetwork.

- A router is similar to other computers in that it has :
  - Memory.
  - Operating system *(In Cisco routers, the operating system is called the Internetwork Operating System, or IOS, and is proprietary to Cisco.)*
  - Storage.
  - User interface.
- What makes a router different from other computers is :
  - The user interface.
  - The configuration of memory.



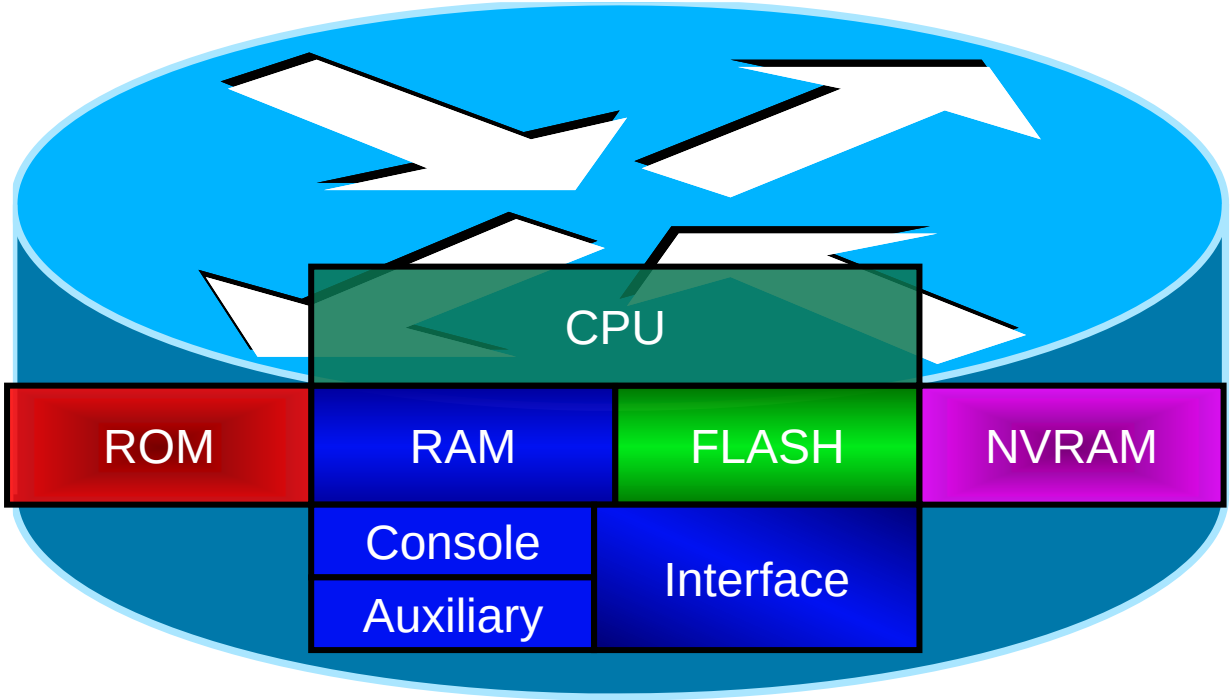
# Router's Component (Cont.)

There are two kinds of router's components:

- Internal components
  - CPU (Central Processing Unit)
  - ROM (Read Only Memory)
  - FLASH
  - NVRAM (Nonvolatile RAM)
  - RAM (Random Access Memory)
- External component (interface)
  - WAN
  - LAN
  - Console/AUX (Auxiliary)

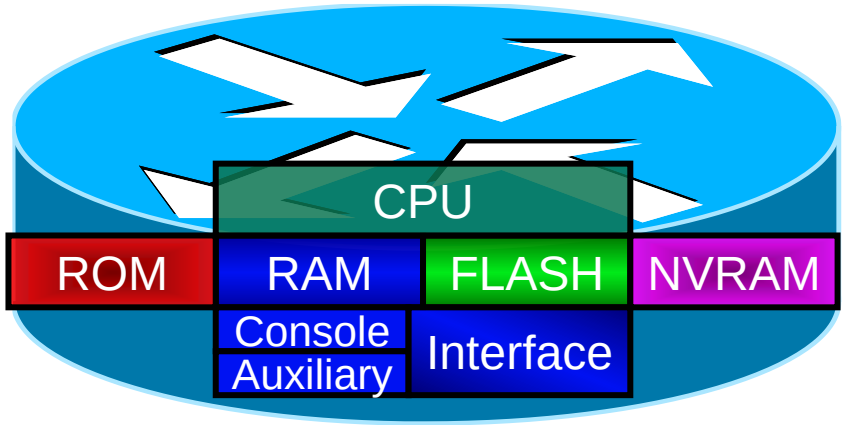


# Router's Component (Cont.)



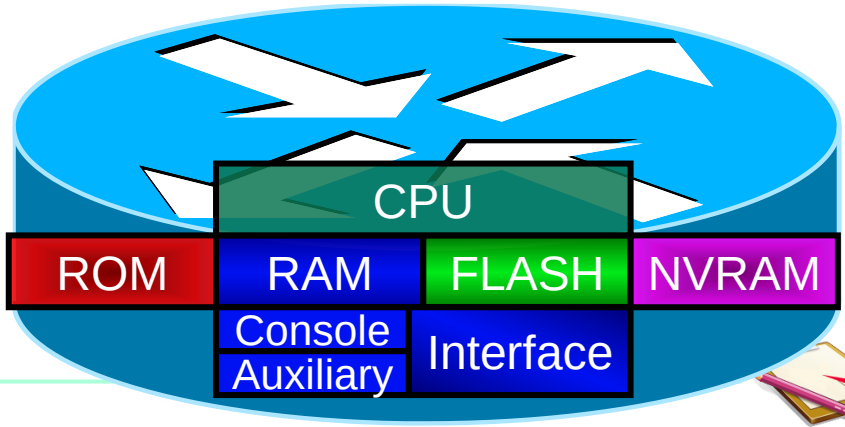
# Internal Component: ROM

- Contains power-on diagnostics, a bootstrap program, and operating system software.
- The main tasks for ROM are hardware diagnostics during router bootup and loading the Cisco IOS software from flash to RAM.
- ROMs are not erasable.
- They can only be upgraded by replacing the ROM chips in the sockets.



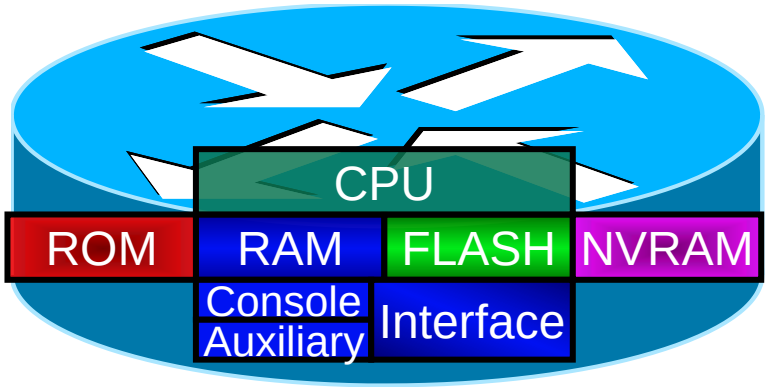
# Internal Component: RAM

- Random Access Memory (RAM) is split by the IOS into Main and Shared memory.
- Main memory is used to store router configuration and IOS data structures such as running-config, routing tables, switching cache,  and ARP tables...
- Shared memory buffers packets waiting to be processed (This type of memory is only used by 2500, 2600, 4000 routers. The 7000 routers have a switch processor that controls the flow of packets through the router).
- The contents of RAM are lost when power is removed.
- RAM can be upgraded.



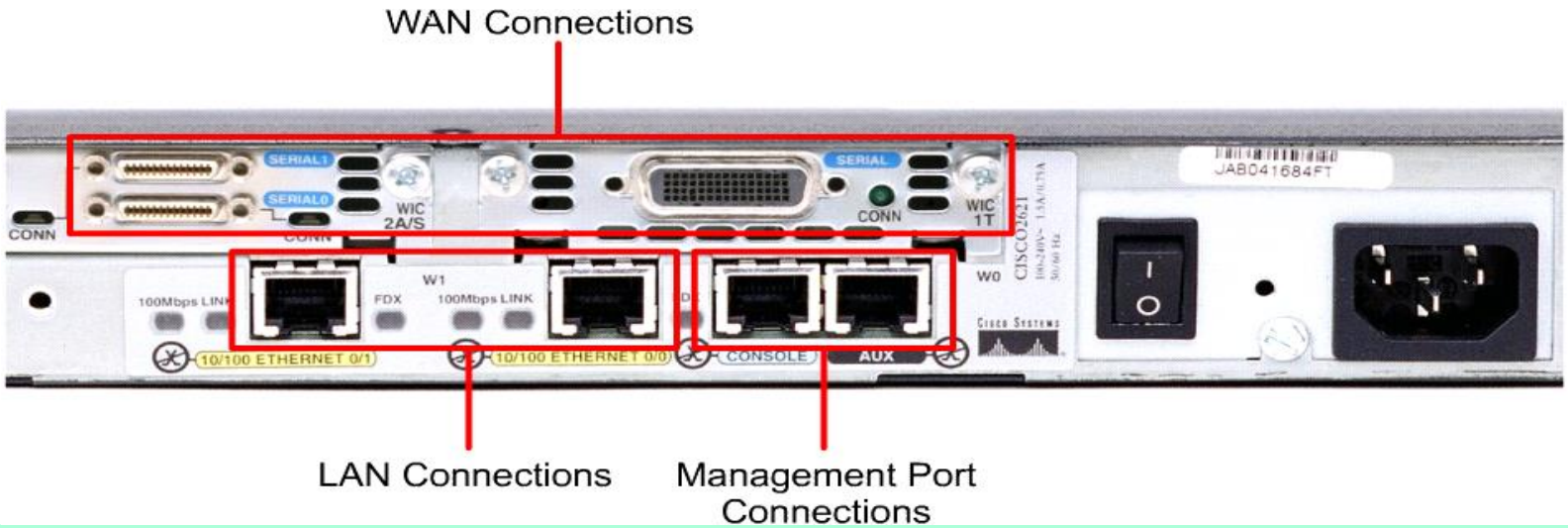
# Internal Component: FLASH

- Flash memory holds the full current version of IOS running on the router.
- Flash memory is erasable memory that can be overwritten with newer versions of the IOS.
- The IOS may be in uncompressed or compressed form.
- In most routers (2600, 3600, 4000,…) an executable copy of the IOS is transferred to RAM during the boot process.
- In other routers such as 2500, the IOS may be run directly from flash.



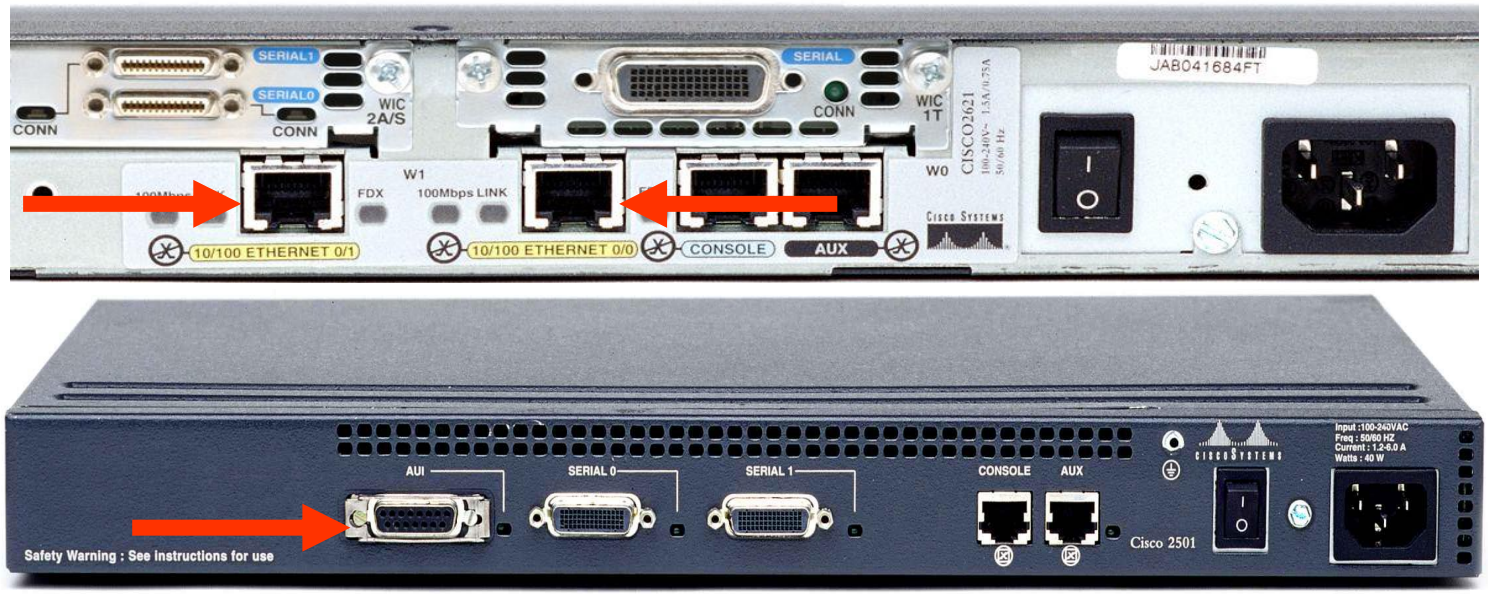
# External Component: Interfaces

- The interfaces are the router connections to the outside.
- The three types of interfaces are:
- Local Area Network (LANs).
- Wide Area Network (WANs).
- Management Ports (Console/AUX).



# External Component: Interfaces (LAN)

- LAN interfaces allow the router to connect to the Local Area Network media.
- This is usually some form of Ethernet.
- However, it could be some other LAN technology such as Token Ring or Asynchronous Transfer Mode (ATM).



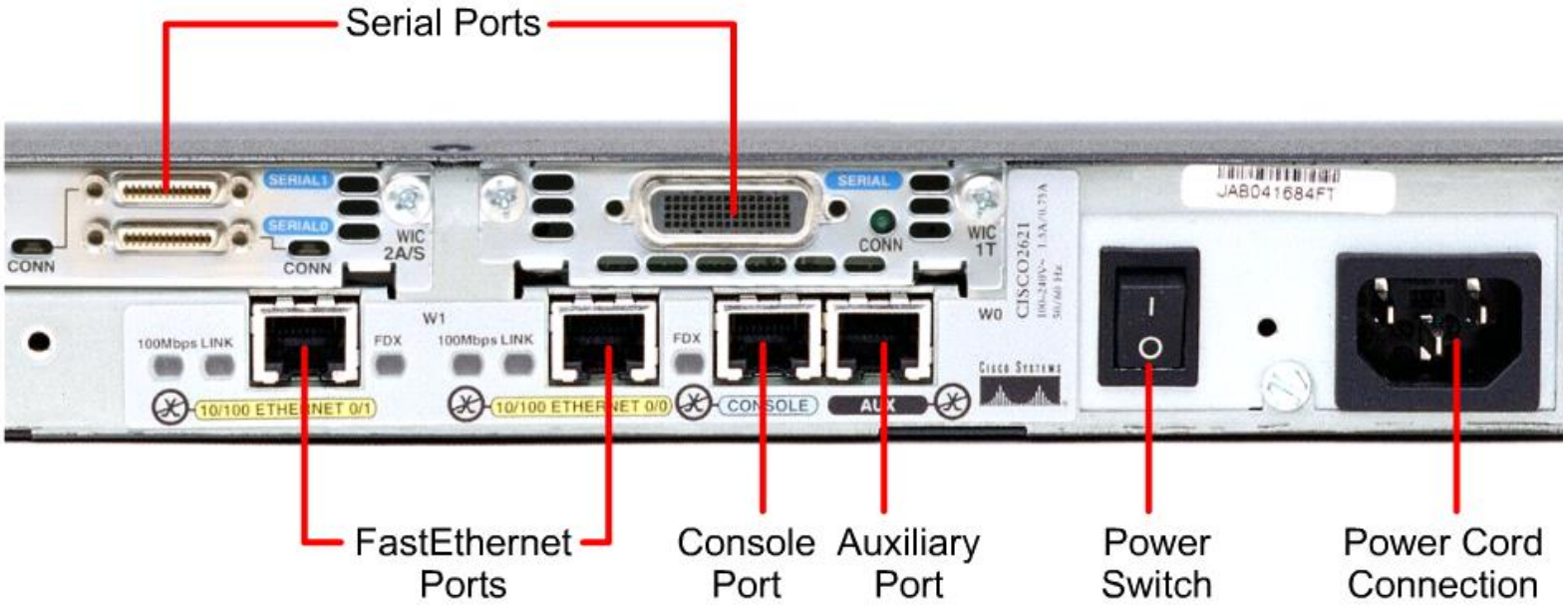
# External Component: Interfaces (Management Port)

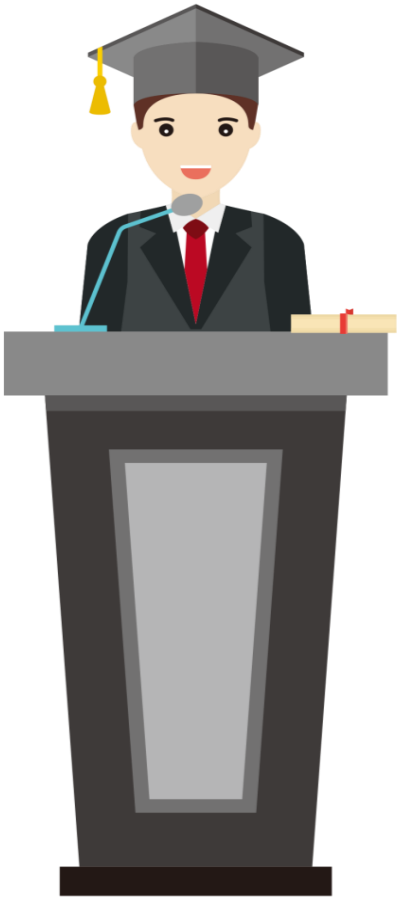
- The function of management ports is different from the other connections.
- The LAN and WAN connections provide network connections through which frame packets are passed.
- The management port provides a text-based connection for the configuration and troubleshooting of the router.
- The common management interfaces are the console and auxilliary ports.



# Router Connections: Console

- The console port is a management port used to provide out-of-band access to the router.
- It is used for the initial configuration of the router, monitoring, and disaster recovery procedures)





# THANK YOU!

